HLS based Hardware Watermarking using an Integrated Framework based on Encoded Dependency Matrix and Encrypted Load-Factor Signature

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• Introduction

- System-on-Chips (SoCs) are integral to consumer electronics (CE) devices such as smart phones, smart watches, set-top boxes, wearable devices etc.
- However, due to globalization in design supply chain [10], IP reuse-based design process is prone to external hardware attacks such from IP seller/vendor perspective such as false IP ownership claim and IP piracy.
- It is crucial to protect the IP vendor (or seller) design from IP piracy and his/her IP ownership rights.

[10] Chang, CH., Potkonjak, M., Zhang, L. (2016). Hardware IP Watermarking and Fingerprinting. In: Chang, CH., Potkonjak, M. (eds) Secure System Design and Trustable Computing. Springer, Cham

• Prior Approaches

Sr. No.	Existing Work	Technique Used	Remark
1.	P. SLPSK et.al., [1] (2023)	SIGNED, a challenge-response along with associated algorithms	However, [1] incurs some design overhead as compare to proposed method
2.	D. Roy et.al., [2] (2019)	Watermarking approach based on binary signatures	[2] produces weaker tamper tolerance ability due to binary encoding process
3.	A. Sengupta et.al.,[5] (2021)	Facial biometric based hardware watermarking	preprocessing steps for watermark generation

[1]. Patanjali SLPSK , A.A. Nair , C. Rebeiro, and S. Bhunia, "SIGNED: A Challenge-Response Scheme for Electronic Hardware Watermarking," *IEEE Trans on Computers*, VOL. 72(6), 2023.

[2] D. Roy et.al, "Multilevel Watermark for Protecting DSP Kernel in CE Systems," *IEEE Consumer Electronics Magazine*, vol. 8, no. 2, pp. 100-102, March 2019.

[5]. A. Sengupta, M. Rathor, "Facial biometric for securing hardware accelerators" *IEEE Trans. Very Large Scale Integr. (VLSI) Syst*. 29(1), 112–123 (2021).

• Threat Model

IP Threat

IP Piracy: Unauthorized copying of IP design under a different brand name by an attacker.

IP Ownership Claim When attacker wrongly claims they created or own original IP.

- ➢ These threats need to be tacked through robust detective countermeasures which are capable to provide definitive proof-of IP authenticity and ownership for an IP vendor in front of a third-party authenticator [9],[10],[11].

[9] S. Akter, K. Khalil and M. Bayoumi, "A Survey on Hardware Security: Current Trends and Challenges," IEEE Access, vol. 11, pp. 77543-77565.

[10] Chang, CH., Potkonjak, M., Zhang, L. (2016). Hardware IP Watermarking and Fingerprinting. In: Chang, CH., Potkonjak, M. (eds) Secure System Design and Trustable Computing. Springer, Cham.

[11] M. Rostami, F. Koushanfar and R. Karri, "A Primer on Hardware Security: Models, Methods, and Metrics," in Proceedings of the IEEE, vol. 102, no. 8, pp. 1283-1295, Aug. 2014.

- # The Proposed Hardware Watermarking Methodology

A. Overview of Proposed Methodology

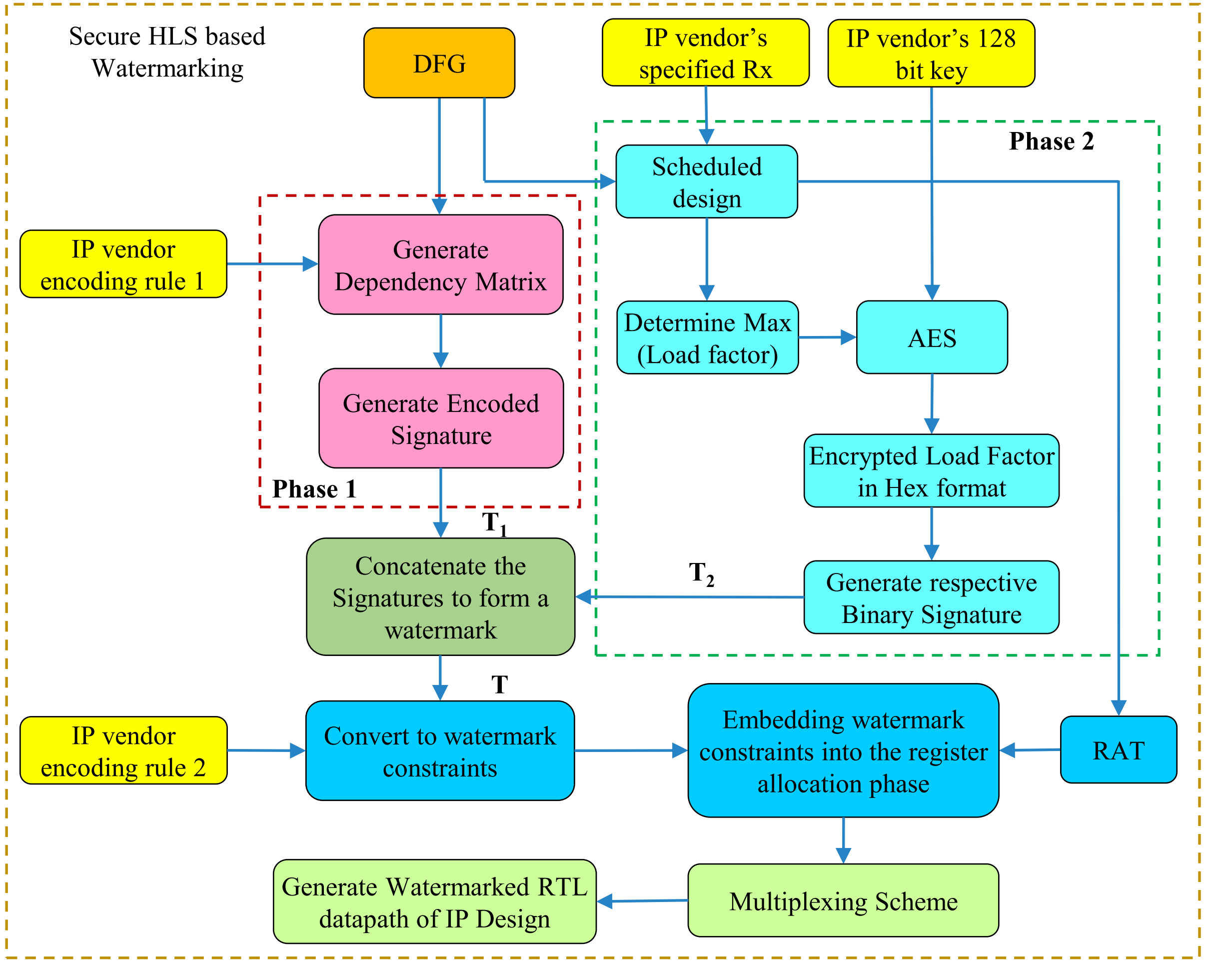


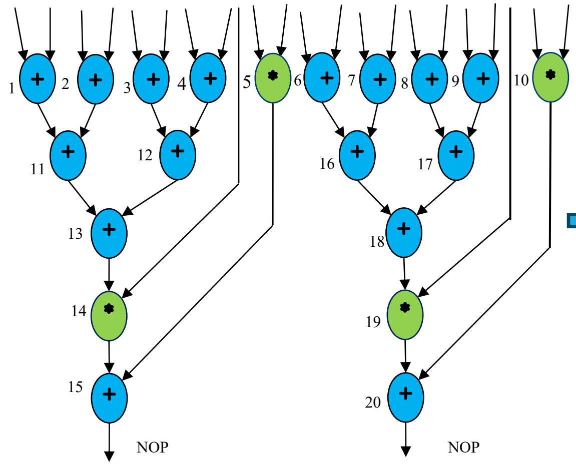
Fig.1. Proposed HLS based Hardware (IP) Watermarking Methodology

• The Proposed Hardware Watermarking Methodology

B. Details of the Proposed Methodology

PHASE - 1

(a). Data flow graph for sharpening filter



(b) Dependency matrix Y_{ixj}

operation	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	P	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
2	0	P	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
3	0	0	P	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
4	0	0	0	P	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
5	0	0	0	0	P	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
6	0	0	0	0	0	P	0	0	0	0	0	0	0	0	0	1	0	0	0	0
7	0	0	0	0	0	0	P	0	0	0	0	0	0	0	0	1	0	0	0	0
8	0	0	0	0	0	0	0	P	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	P	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	P	0	0	0	0	0	0	0	0	0	1
11	1	1	0	0	0	0	0	0	0	0	P	0	1	0	0	0	0	0	0	0
12	0	0	1	1	0	0	0	0	0	0	0	P	1	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	1	1	P	1	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	1	P	1	0	0	0	0	0
15	0	0	0	0	1	0	0	0	0	0	0	0	0	1	P	0	0	0	0	0
16	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	P	0	1	0	0
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	P	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	P	1	0
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	P	1
20	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	P

(c) Signature (400 digits) T1

P000000000100000000000P00000000010000000
0000P000000000100000000000P000000010000
00000000P000000000010000000000P00000000
010000000000P00000000100000000000P0000
0000000000000000P0000000000000000000000P
00000000011100000000P010000000001100000
00P1000000000000000000011P10000000000000
000001P100000000010000000001P0000000000
1100000000P0100000000000000000000P100000
00000000000011P1000000000000000000001P10
0000000001000000001P

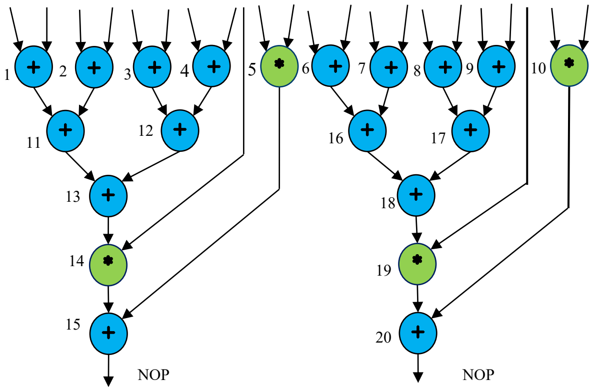
Fig. 2 (A). Flow Diagram of proposed secured watermark approach

• The Proposed Hardware Watermarking Methodology

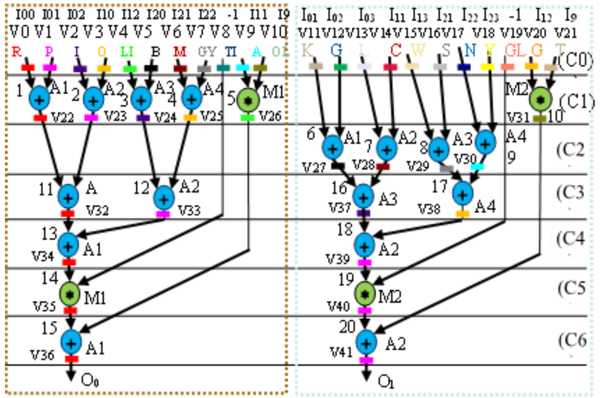
B. Details of the Proposed Methodology

PHASE - 2

(a). Data flow graph for sharpening filter



(d)



Scheduled data flow graph for sharpening filter using 2M and 4A

(e)

Operation #	Load factor ($LF^{op'n'}$)
1	$LF^{OP1} = 529.94$
2	$LF^{OP2} = 529.94$
3	$LF^{OP3} = 529.94$
4	$LF^{OP4} = 529.94$
5	$LF^{OP5} = 331.21$
6	$LF^{OP6} = 529.94$
7	$LF^{OP7} = 529.94$
8	$LF^{OP8} = 529.94$
9	$LF^{OP9} = 529.94$
10	$LF^{OP10} = 331.21$
MAX [$LF^{OP1}, LF^{OP2}, \ldots, LF^{OP10}$] = 529.94	

(f)

IP vendor's 128 bits key

Post AES, Hex Code (T_2)
e63e076be993de0a7e501fba484580b4

Hex to Binary Code (128 bits)
T_2
11100110001111100000001110110101111101001100100111101111000001010011111100101000000011111101110100100100000100010110000000010110100

Concatenation of T1||T2

Output of phase 1
T1

(g) Final template 528 digits (T)

P0000000001000000000P000000001000000000000P00000000100000000000P0000000100000000000000P00000000010000000000P0000000001000000000P000000001000000000000P00000000000000000000000P000000000000000000000P0000000001110000000000P010000000000110000000P10000000000000000000011P1000000000000000000001P1000000000010000000001P00000000000110000000P010000000000000000000P1000000000000000000011P1000000000000000000001P10000000001000000001P11100110001111100000001110110101111101001100100111101111000001010011111100101000000011111101110100100100000100010110000000010110100.

Encoding Rule 2

(h) Converting into watermarking constraints

Constraints corresponding to watermark signature bit '1': (V1,V3),(V1,V5),……….,(V11,V21).

Constraints corresponding to watermark signature bit '0': (V0,V4),(V0,V6),……, (V38,V40).

Constraints corresponding to watermark signature digit 'P': (V0,V2),(V0,V3),………, (V0,V71).

Fig.2 (B). Flow Diagram of proposed secured watermark approach

The Proposed Hardware Watermarking Methodology

Table 1. RAT pre-embedding hardware security constraints corresponding to sharpening filter

Reg.\C.S.	CO	C1	C2	C3	C4	C5	C6
Red (R)	V0	V22	V22	V32	V34	V35	V36
Purple (P)	V1	V23	V23	V33	V39	V40	V41
Indigo (I)	V2	V24	V24	V37	-	-	-
Orange (O)	V3	V25	V25	V38	-	-	-
Lime (LI)	V4	V26	V26	V26	V26	V26	
Blue (B)	V5	-	V27	-	-	-	-
Maroon(M)	V6	-	V28	-	-	-	-
Gray (GY)	V7	-	V29	-	-	-	-
Teal (TL)	V8	V8	V8	V8	V8	-	-
Aqua (A)	V9	-	V30	-	-	-	-
Olive (OL)	V10	V31	V31	V31	V31	V31	-
Khaki (K)	V11	V11	-	-	-	-	-
Green (G)	V12	V12	-	-	-	-	-
Lavender (L)	V13	V13	-	-	-	-	-
Crimson(C)	V14	V14	-	-	-	-	-
Wheat (W)	V15	V15	-	-	-	-	-
Silver (S)	V16	V16	-	-	-	-	-
Navy (N)	V17	V17	-	-	-	-	-
Yellow(Y)	V18	V18	-	-	-	-	-
Gold (GL)	V19	V19	V19	V19	V19	-	-
Grey (G)	V20	-	-	-	-	-	-
Tan (T)	V21	-	-	-	-	-	-

Table 2. RAT post- embedding hardware security constraints corresponding to sharpening filter

Reg.\C.S.	CO	C1	C2	C3	C4	C5	C6
Red (R)	V0	~~V22~~	~~V22~~	~~V32~~	~~V34~~	~~V35~~	~~V36~~
Purple (P)	V1	~~V23~~	~~V23~~	~~V33~~	~~V39~~/V34	~~V40~~	~~V41~~
Indigo (I)	V2	~~V24~~/V25	~~V24~~/V25	~~V37~~	-	-	-
Orange (O)	V3	~~V25~~/V24	~~V25~~/V24	~~V38~~	-	-	-
Lime (LI)	V4	~~V26~~	~~V26~~	~~V26~~	~~V26~~	~~V26~~	V41
Blue (B)	V5	-	~~V27~~	V32	-	-	-
Maroon(M)	V6	V23	~~V28~~/V23	-	-	-	-
Gray (GY)	V7	-	~~V29~~/V28	-	-	-	-
Teal (TL)	V8	V8	V8	V8	V8	V35	-
Aqua (A)	V9	V26	~~V30~~/V26	V26	V26	V26	-
Olive (OL)	V10	V31	V31	V31	V31	V31	-
Khaki (K)	V11	V11	-	-	-	-	V36
Green (G)	V12	V12	-	V33	-	-	-
Lavender (L)	V13	V13	-	V38	-		-
Crimson(C)	V14	V14	-	-	V39	-	-
Wheat (W)	V15	V15	-	-	-	-	
Silver (S)	V16	V16	-	V37	-	-	-
Navy (N)	V17	V17	V30	-	-	-	-
Yellow(Y)	V18	V18	V29	-	-	-	-
Gold (GL)	V19	V19	V19	V19	V19	V40	-
Grey (G)	V20	-	V27	-	-	-	-
Tan (T)	V21	V22	V22	-	-	-	-

• Watermark Detection/Validation

An attacker needs to decode the following successfully to falsely establish IP watermark:

1. The IP vendor's specified resource constraints (Rx) used for generating SDFG.
2. Dependency matrix that is based on IP vendor encoding rule-1.
3. Encoded signature (T1).
4. Load factor maximum value.
5. AES encryption with an IP vendor specified 128-bit key.
6. Signature (T2).
7. The concatenation order to form the final signature template (T1||T2=T).
8. IP vendor encoding rule 2 used for generating the watermark constraints.

• Results and Analysis

- ***Security Analysis*** - From the literatures [3], [5], two standard security metrics have been used to evaluate the security of the proposed approach: probability of coincidence (Ci) and tamper tolerance (TA).
  - Ci measures the likelihood of an unsecured IP design coincidentally containing the original watermark constraints (digital evidence). Ci is represented by eqn. (1) :

$$Ci = (1 - 1/Q)^{v} \quad (1)$$

  - Further, tamper tolerance measures the tolerance ability of a watermarking approach against tampering attack. TA is represented by eqn. (2) :

$$TA = S^{w} \quad (2)$$

- ***Watermark Design Cost Analysis***- The watermark design cost is estimated using eqn. (3) [5]:

$$Design\ cost = x1(Ar/Amax) + x2(Lc/Lmax) \quad (3)$$

[3]. F. Koushanfar, I. Hong, and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection,"*ACM Trans*. Design Autom. Electron. Syst., vol.10(3), pp. 523–545, Jul. 2005.
[5]. A. Sengupta, M. Rathor, "Facial biometric for securing hardware accelerators" *IEEE Trans. Very Large Scale Integr. (VLSI) Syst*. 29(1), 112–123 (2021).

• Results and Analysis

Table 3. Comparison of achieved C_i through the proposed approach with other related techniques for blur filter IP design

Approaches	Number of unique colors/ distinct registers	Embedded Watermark Constraints	C_i
Proposed Approach	20	283	4.96E-7
Pragma based watermarking, 2021 [4]		71	2.62E-2
FSM watermarking, 2022 [7]		128	1.40E-3
Facial biometric, 2021 [5]		83	1.41E-2
DNA biometric, 2022 [8]		128	1.40E-3
Automated watermarking,2008 [6]		160	2.72E-4

Table 4. Comparison of achieved Ci through the proposed approach with other related techniques for Sharpening filter IP design

Approaches	Number of unique colors/ distinct registers	Embedded Watermark Constraints	C_i
Proposed Approach	22	324	2.84E-7
Pragma based watermarking, 2021 [4]		71	3.67E-2
FSM watermarking, 2022 [7]		128	2.59E-3
Facial biometric, 2021 5]		83	2.10E-2
DNA biometric, 2022 [8]		128	2.59E-3
Automated watermarking, 2008 [6]		160	5.85E-4

Table 5. Comparison of achieved Ci through the proposed approach with other related techniques for FIR filter IP design

Approaches	Number of unique colors/ distinct registers	Embedded Watermark Constraints	C_i
Proposed Approach	21	331	9.69E-8
Pragma based watermarking, 2021 [4]		71	3.13E-2
FSM watermarking, 2022 [7]		128	1.93E-3
Facial biometric, 2021 [5]		83	1.74E-2
DNA biometric, 2022 [8]		128	1.93E-3
Automated watermarking,2008 [6]		160	4.07E-4

Table 6. Comparison of achieved Ci through the proposed approach with other related techniques for Laplace filter IP design

Approaches	Number of unique colors/distinct registers	Embedded Watermark Constraints	C_i
Proposed Approach	14	167	4.21E-6
Pragma based watermarking, 2021 [4]		71	5.18E-3
FSM watermarking, 2022 [7]		128	7.59E-5
Facial biometric, 2021 [5]		83	2.13E-3
DNA biometric, 2022 [8]		128	7.59E-5
Automated watermarking,2008 [6]		160	7.08E-6

• Results and Analysis

Table 7. Comparison of achieved T_A through the proposed approach with other related techniques

Approaches	Blur Filter		Sharpening Filter		FIR Filter		Laplace Filter	
	Watermark strength (digits)	T_A	Watermark strength (digits)	T_A	Watermark strength (digits)	T_A	Watermark strength (digits)	T_A
Proposed Approach	452	4.55E+215	528	8.31E+251	528	8.31E+251	272	5.98E+129
Automated watermarking,2008 [6]	160	1.46E+48	160	1.46E+48	160	1.46E+48	160	1.46E+48
Pragma based watermarking, 2021 [4]	71	NA	71	NA	71	NA	71	NA
FSM watermarking, 2022 [7]	128	3.40E+38	128	3.40E+38	128	3.40E+38	128	3.40E+38
Facial biometric, 2021 [5]	83	9.67E+24	83	9.67E+24	83	9.67E+24	83	9.67E+24
DNA biometric, 2022 [8]	128	3.04E+38	128	3.04E+38	128	3.04E+38	128	3.04E+38

Table 8. Cost Overhead analysis of Proposed Approach

Benchmarks	Pre-Watermarked Embedded Design			Post-Watermarked Embedded Design			Cost Overhead (%)
	Area (μm^2)	Latency (ps)	Cost	Area (μm^2)	Latency (ps)	Cost	
FIR Filter	186.38	1722.31	0.43	186.38	1722.31	0.43	0
Sharpening Filter	243.62	794.9	0.58	243.62	794.9	0.58	0
Blur Filter	109.96	1523.54	0.67	109.96	1523.54	0.67	0
Laplace Filter	199.64	728.66	0.68	199.64	728.66	0.68	0

• References

[1]. P. SLPSK , A.A. Nair , C. Rebeiro, and S. Bhunia, "SIGNED: A Challenge-Response Scheme for Electronic Hardware Watermarking," *IEEE Trans on Computers*, VOL. 72(6), 2023.

[2] D. Roy et.al, "Multilevel Watermark for Protecting DSP Kernel in CE Systems," *IEEE Consumer Electronics Magazin*e, vol. 8, no. 2, pp. 100-102, March 2019.

[3] F. Koushanfar, I. Hong, and M. Potkonjak, ‘‘Behavioral synthesis techniques for intellectual property protection,’’*ACM Trans*. Design Autom. Electron. Syst., vol.10(3), pp. 523–545, Jul. 2005.

[4] J. Chen, & B. C. Schafer, “Watermarking of behavioral IPs: A practical approach” in 2021 *Design, Automation & Test in Europe Conference & Exhibition (DATE), France*, pp. 1266–1271.

[5] A. Sengupta, M. Rathor, “Facial biometric for securing hardware accelerators” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst*. 29(1), 112–123 (2021).

[6] E. Castillo, et al. “Automated signature insertion in combinational logic patterns for HDL IP core protection” in 2008 4th Southern Conference on Programmable Logic, Bariloche, Argentina, pp. 183–186 (2008).

[7] R. Karmakar, S. S. Jana and S. Chattopadhyay, “A Cellular Automata Guided Finite-State-Machine Watermarking Strategy for IP Protection of Sequential Circuits,” *IEEE Trans. Emerg. Topics Comput*., vol. 10, no. 2, pp. 806-823, 1 April-June 2022.

[8] A. Sengupta, R. Chaurasia, “Securing IP cores for DSP applications using structural obfuscation and chromosomal DNA impression” *IEEE Access.*10, 50903–50913 (2022).

[9] S. Akter, K. Khalil and M. Bayoumi, "A Survey on Hardware Security: Current Trends and Challenges," *IEEE Access*, vol. 11, pp. 77543-77565.

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[12] NanGate 15 nm Open Cell Library. [Online]. http://www.nangate.com/?pageid=2328, Accessed March 2025

[13] ExPRESS – Benchmarks, University of California. Santa Barbara, https://web.ece.ucsb.edu/EXPRESS/benchmark/, Last accessed on March 2025.

[14] A. Sengupta and R. Sedaghat, "Integrated scheduling, allocation and binding in High Level Synthesis using multi structure genetic algorithm based design space exploration," *12th IEEE International Symposium on Quality Electronic Design*, CA, USA, 2011, pp. 1-9.

THANK YOU